Test 8

Dr G. Scarciotti

February 27, 2025

- **Please submit a pdf with your answers on blackboard by Thursday 6th of March at 13:00.**

  Use the handouts to reply to these questions. Do not use generative AI tools. Note that each answer is in the handouts.

  You cannot answer using screenshots. You need to reply by typing or in your own writing. This promotes subconscious retention.

  This is an individual test. You cannot collaborate with other students or any other individual. This must be your work.

  You should be able to complete this in 1 page. Do not write, nor spend too much time on this.

Please state name, surname and CID.

1. (3 marks) Provide the barrier function. Explain how the barrier function is used to approximate the problem with the indicator function I_-. Explain what happens when one of the inequality constraints approaches zero.

2. (2 marks) Provide the centrality condition. Consider an LP in inequality form and the hyperspace defined by its inequality constraints. For such a problem, sketch an example of centrality path and explain it.

3. (2 marks) Give the modified complementary condition associated to the centrality condition and explain it as a function of t.

4. (1 marks) Explain the differences between the two Phase I methods seen in the lecture.

5. (2 marks) Provide the duality gap associated to a convex optimisation problem with m generalised inequalities which are K_i-convex and have generalised logarithms with degree θ_i. Explain the role of θ_i in the computational complexity of the barrier method.

Non-assessed questions, for self-study - Do not submit

- Explain why we cannot use the indicator function I_- in convex optimisation.
- For a given central point $x^*(t)$, provide the dual variables $\lambda^*(t)$ and $\mu^*(t)$. Provide and explain the duality gap associated to $x^*(t)$, $\lambda^*(t)$ and $\mu^*(t)$.
- Provide the barrier algorithm and explain it. In particular, define the inner and the outer iterations.

- Explain the role of μ and the trade-off that this parameter plays with respect to the number of inner and outer iterations.
- Assume self-concordance of all centering steps. Provide an upper bound on the total number of Newton steps required by the barrier method (omit the complexity of the initial centering step).